0.1 Fermionic systems with determinants

In this section we review section 5.6 of the book *Quantum Monte Carlo Approaches for Correlated Systems* by Federico Becca and Sandro Sorella. The presentation closely follows Becca and Sorella, with additional details included for clarity.

Throughout this section, we assume a lattice that contains L sites and each site can contain one spin-up and one spin-down orbital. Therefore, we have $2L$ available orbitals that can be occupied by particles. We use one collective index that contains the site and the spin, by setting

$$\hat{c}_{i,\uparrow} \equiv \hat{d}_i \quad \text{and} \quad \hat{c}_{i,\downarrow} \equiv \hat{d}_{i+L}. \tag{1}$$

This corresponds to the fermionic-mode ordering $(1\uparrow, 2\uparrow, \ldots, L\uparrow, 1\downarrow, 2\downarrow, \ldots, L\downarrow)$. Any fixed ordering is fine, but it must be used consistently throughout.

Any number-conserving quadratic fermionic Hamiltonian can be diagonalized:

$$\hat{H}_0 = \sum_{I,J=1}^{2L} t_{I,J} d_I^\dagger \hat{d}_J \tag{2}$$

$$= \boldsymbol{d}^\dagger \boldsymbol{T} \boldsymbol{d} = \boldsymbol{d}^\dagger \boldsymbol{U}\boldsymbol{U}^\dagger \boldsymbol{T} \boldsymbol{U}\boldsymbol{U}^\dagger \boldsymbol{d} \tag{3}$$

$$= \boldsymbol{\phi}^\dagger \boldsymbol{E} \boldsymbol{\phi} = \sum_{\alpha=1}^{2L} \varepsilon_\alpha \hat{\Phi}_\alpha^\dagger \hat{\Phi}_\alpha, \tag{4}$$

$$\text{with} \quad \phi_\alpha^\dagger = \sum_I U_{I,\alpha} d_I^\dagger. \tag{5}$$

The ground state is thus obtained by occupying the N_e lowest-energy single-particle orbitals:

$$|\Phi_0\rangle = \prod_{\alpha=1}^{N_e} \hat{\Phi}_\alpha^\dagger |0\rangle = \prod_{\alpha=1}^{N_e} \left(\sum_{I=1}^{2L} U_{I\alpha} \hat{d}_I^\dagger \right) |0\rangle . \tag{6}$$

The matrix U is in general a $2L \times 2L$ unitary matrix, where we only need the first N_e columns to construct the ground state, as can be seen above.

A general occupation-number configuration can be written as

$$|x\rangle = \hat{d}_{R_1}^\dagger \hat{d}_{R_2}^\dagger \cdots \hat{d}_{R_{N_e}}^\dagger |0\rangle , \tag{7}$$

where $R_1, \ldots, R_{N_e}$ are the occupied orbitals in the configuration x, with $R_j = 1, \ldots 2L$. This is of course a redundant labeling, since for example $c_1^\dagger c_2^\dagger |0\rangle = -c_2^\dagger c_1^\dagger |0\rangle$. But to describe arbitrary configurations, that can occur from applying the Hamiltonian to another arbitrary configuration, they are convenient. The resulting operator string need not be in canonical order and it then must be reordered, producing the appropriate fermionic sign. Thus, different orderings may represent the same physical configuration up to a sign. Therefore, we have

$$\begin{aligned}\langle x|y\rangle &= \langle 0|\, \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \hat{d}^\dagger_{Q_1} \cdots \hat{d}^\dagger_{Q_{N_e}} |0\rangle \\ &= \begin{cases} 0, & \text{iff } \{R_1, \ldots, R_{N_e}\} \neq \{Q_1, \ldots, Q_{N_e}\} \\ \text{sign}(p), & \text{else, with } p\left((Q_1, \ldots, Q_{N_e})\right) = (R_1, \ldots, R_{N_e}), \end{cases} \end{aligned} \tag{8}$$

where $\{.\}$ denotes a mathematical set and $(.)$ an ordered tuple.

The overlap between a general configuration $|x\rangle$ and the ground state of the quadratic Hamiltonian is thus given as

$$\begin{aligned}\langle x|\Phi_0\rangle &= \langle 0|\, \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \prod_{\alpha=1}^{N_e} \left(\sum_{i=1}^{2L} U_{i\alpha} \hat{d}^\dagger_i \right) |0\rangle \\ &= \langle 0|\, \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \left[\sum_{p \in P} \text{sign}(p) \prod_{\alpha=1}^{N_e} U_{p\{Q_j\},\alpha} \right] \hat{d}^\dagger_{Q_1} \cdots \hat{d}^\dagger_{Q_{N_e}} |0\rangle \,, \end{aligned} \tag{9}$$

with P containing all possible permutations of the Q_j. These have now to be restricted on all valid permutations of the R_j within x (see eq. (8)), which yields

$$\langle x|\Phi_0\rangle = \langle 0|\, \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \left[\sum_{p \in P} \text{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha} \right] \hat{d}^\dagger_{R_1} \cdots \hat{d}^\dagger_{R_{N_e}} |0\rangle \tag{10}$$

$$= \sum_{p \in P} \text{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha} \tag{11}$$

$$= \det \left[U_{R_i,\alpha} \right]_{\substack{i=1,\ldots,N_e \\ \alpha=1,\ldots,N_e}} , \tag{12}$$

where we used the Leibniz formula in the last step.

In other words, although U is a $2L \times 2L$ matrix, only the N_e columns corresponding to the occupied lowest-energy orbitals and the N_e rows corresponding to the occupied orbitals R_i of the configuration x are needed:

$$U_x = \begin{pmatrix} U_{R_1,1} & \cdots & U_{R_1,N_e} \\ \vdots & \ddots & \vdots \\ U_{R_{N_e},1} & \cdots & U_{R_{N_e},N_e} \end{pmatrix} \tag{13}$$

and hence

$$\langle x|\Phi_0\rangle = \det(U_x). \tag{14}$$

This construction can also be used for BCS-type Hamiltonians (see end of section 5.6.1 of the book by Becca and Sorella).

Variational Monte Carlo

Let's say we have some general fermionic Hamiltonian $\hat{H}$, which might not be quadratic and we choose the Slater determinant as our variational ansatz $\langle x|\Phi_U\rangle = \det(U_x)$ with the entries of the full U as the variational parameters. The variational Monte Carlo (VMC) workflow then roughly looks like

1. Sample fermionic occupation configurations $x = (R_1, \ldots, R_{N_e})$ following the distribution resulting from $\Phi_U(x) = \det(U_x)$.
2. Evaluate Hamiltonian matrix elements $\langle x|\,\hat{H}\,|x'\rangle$ and estimate the local energy $e_L(x) = \sum_{x'} \langle x|\,\hat{H}\,|x'\rangle \frac{\langle x'|\Phi_U\rangle}{\langle x|\Phi_U\rangle}$.
3. Compute the gradient of the energy $\langle E\rangle \approx \frac{1}{N}\sum_{n=1}^{N} e_L(x_n)$ and optimize the variational parameters with it.
4. Repeat until convergence.

The Jordan–Wigner mapping

The Jordan–Wigner mapping (JWM) does not change anything in the derivation above since it is only a rewriting of the operators (and besides doing the above derivation with the JWM would probably be much more tedious). But it might come in handy when computing matrix elements such as $\langle x|\,\hat{H}\,|x'\rangle$. To see what changes in the VMC workflow described above, we only have to look at how the form of the ansatz $\Phi_U(x)$ might change, since we know that the fermionic and spin configurations map onto each other (see **??**). The overlap written with fermionic operators is given as (see eq. (10))

$$\langle x|\Phi_0\rangle_f = \left[\sum_{p\in P} \operatorname{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha}\right] \langle 0|\,\hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \hat{d}^\dagger_{R_1} \cdots \hat{d}^\dagger_{R_{N_e}}\,|0\rangle\,.$$

Using the JWM on a fermionic basis state $|x_b\rangle_f$ maps it to a spin basis state $|x_b\rangle_s$ (see **??**). Thus, applying the JWM to some fermionic configuration state $|x\rangle_f = \operatorname{sign}(x_f)\,|x_b\rangle_f$ maps it to some spin configuration state $|x\rangle_s = \operatorname{sign}(x_s)\,|x_b\rangle_s$. Therefore, we have $\langle x|x\rangle_f = \langle x|x\rangle_s = 1$ and

$$\begin{aligned}\langle x|\Phi_0\rangle_f &= \det U_x \,\langle x|x\rangle_f \\ &= \det U_x \,\langle x|x\rangle_s \\ &= \langle x|\Phi_0\rangle_s\end{aligned}$$

and thus the computation of the overlap does not change. **Hence, the only significant change in the VMC workflow is that the Hamiltonian has to be rewritten but the ansatz remains unchanged.**

References

- *Quantum Monte Carlo Approaches for Correlated Systems* by Federico Becca and Sandro Sorella
- *Fermions and Jordan–Wigner String* – ITensor C++ Documentation